H5.

- The assignment is due at Gradescope on 4/25/25.
- A LaTeX template will be provided for each homework. You are strongly encouraged to type your homework into this template using LaTeX. If you are writing by hand, please fill in the solutions in this template, inserting additional sheets as necessary. This will help facilitate the grading.
- You are permitted to discuss the problems with up to 2 other students in the class (per problem); however, *you must write up your own solutions, in your own words*. Do not submit anything you cannot explain. If you do collaborate with any of the other students on any problem, please list all your collaborators in the appropriate spaces.
- Similarly, please list any other source you have used for each problem, including other textbooks or websites.
- *Show your work.* Answers without justification will be given little credit.
- Your homework is *resubmittable*. Please refer to the course syllabus on Canvas for a more detailed description of this. For any problem that you have not changed from your last submission, please make sure to indicate this in your submission to help our graders grade faster.
- Each student can request a total of up to 5 days of extensions on all homework - either for original submission or resubmission. Up to two days can be requested for a homework. If you would like to request an extension, please fill out the Homework extension form. Extensions do not carry any late penalty. If there is a life emergency that might prevent you from submitting on time not covered by this policy, email the Head TA.
- Is anyone still reading these?

Problem 1 (Short Answer) *For each of the following questions, select true or false, or fill in the correct answer.* ***Please provide a short explanation for your answer.***

(a) *A directed graph is a DAG if and only if it is linearizable, i.e. has a topological sort.*

True ◯ *False* ◯

(b) *Given any matrix* $\mathbf{X}$, *computing* $\mathbf{X}^n$ *using only matrix multiplications and additions requires at least* n *matrix multiplications.*

True ◯ *False* ◯

(c) *For a directed graph* G, *if DFS calls* `previsit(v)` *on all nodes before calling* `postvisit(v)` *any node, then* G *cannot have a cycle.*

True ◯ *False* ◯

(d) *For an execution of DFS on a directed graph* G *with* n *vertices, if every call to* `explore(v)` *terminates immediately (i.e. it does not visit any new node), then* G *must have* n *strongly connected components.*

True ◯ *False* ◯

(e) *If the output of the FFT is* $\begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix}$, *then the input must also have been* $\begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix}$.

True ◯ *False* ◯

(f) *Suppose we want to use FFT to multiply* $x^6 + 3x^5 - 7x^3 + 10$ *with* $x^3 - 2x^2 + x - 4$. *What is the smallest power of 2 for which the FFT matrix of that dimension will successfully evaluate and interpolate the polynomials?*

(g) *What is the output of FFT when the input is* $\begin{pmatrix} 0 & 0 & 1 & 0 \end{pmatrix}$*?*

(h) *Removing an edge in an undirected graph with n nodes and m edges could increase the number of* ***connected components*** *by at most how many components?*

(i) *Removing an edge in a directed graph with n nodes and m edges could increase the number of* ***strongly connected components*** *by at most how many components?*

Problem 2 (RGB trail) *You are given a directed graph $G = (V, E)$, where each edge is colored either red, green, or blue. A trail in G is called an RGB trail if its sequence of edge colors is red, green, blue, red, green, blue, and so on. More formally, a trail $v_0, v_1, \ldots, v_k$ is an RGB trail if, for every integer $i \geq 0$, the edge (v_i, v_{i+1}) is red if $i \pmod 3 = 0$, white if $i \pmod 3 = 1$, and blue if $i \pmod 3 = 2$. Edge colors can be accessed using a function* `COLOR(e)` *which takes a single edge $e \in E$ as input and returns either* `RED`, `GREEN`, *or* `BLUE`. *Give an algorithm to find all vertices v in G that can be reached from a given vertex s through an RGB trail. Your algorithm should be as efficient as possible.*

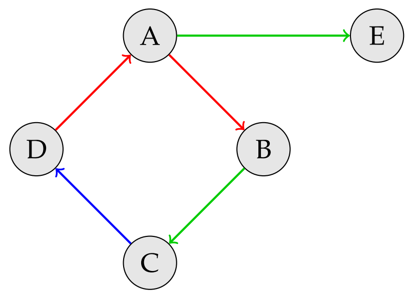

Figure 1: In order to reach E from A, we need to take the RGB trail (A, B, C, D, A, E)

Remember that a trail can visit a vertex more than once, but can only use each edge at most once. As we can see in the above example, the RGB trail from A to E visits A twice.

(a) *Write **pseudocode** for your algorithm.*

(b) *Prove that your algorithm is correct.*

(c) *Analyze the running time of your algorithm.*

(d) *Give a reduction to DFS. (Reducing problem A to problem B means that we can transform instances of problem A into instances of problem B in a way such that solving the instance for problem B gives the solution to the original instance of problem A. For this problem, we are given a directed graph $G = (V, E)$ where each edge is colored either red, green, or blue, together with a source vertex s. This is the original instance of the RGB trail problem. Reducing to DFS means that you need to construct a new graph $G' = (V', E')$ together with assignments of edge colors such that running DFS on G' gives you the solution to the original instance of the RGB trail problem.)*